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Enabling reduced-order collisional-radiative modeling
for reactive plasma physics simulations

Richard J. E. Abrantes

National Research Council - Postdoctoral Research Associate
In-Space Propulsion Branch (RQRS)

UCLA Space Institute Seminar, Nov. 30th, 2021
Los Angeles, California, USA

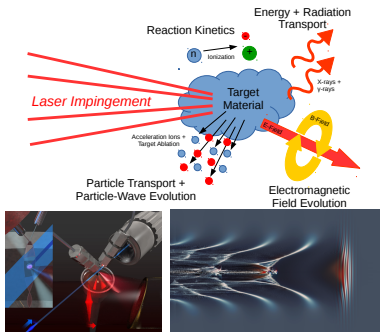
Introduction

Objective:

- ⇒ Explore physical models and techniques that can produce accurate and more efficient simulations for our plasma applications of interest

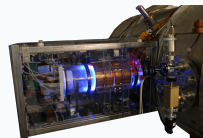
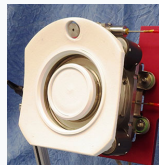
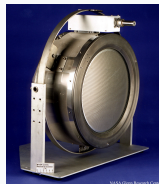
Laser-Plasma Concepts and Devices

(AFRL Wright Patterson AFB)



Electric Propulsion Advancement

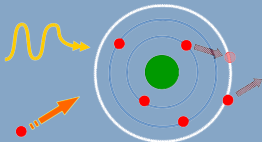
(AFRL Edwards AFB)



Overview: Collisional-Radiative Model

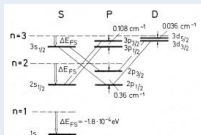
Physical description

- ▶ Atomic states
- ▶ Electron-impact kinetics
- ▶ Radiation-based kinetics
- ▶ Can be supplied by atomic models



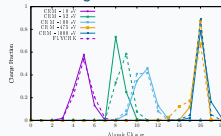
Numerical calculations

- ▶ Solving for atomic state distribution (ASD)
- ▶ Steady-state populations / Time-dependent evolution
 - ▶ Stiff, nonlinear coupled system!

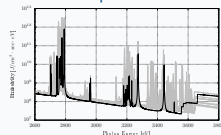


Simulated Plasma

Charge Distribution



Spectra



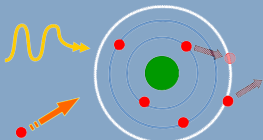
Challenges:

- ▶ Enumeration of atomic states and transitions (+ types of transitions!)
- ▶ Collisional-radiative effects are inherently connected to other physics modules

Features of ACRE Collisional-Radiative Model

Atomic Models

- ▶ Los Alamos (LANL) Atomic Physics Codes
- ▶ cFAC, forked C++ version of Flexible Atomic Code (FAC)

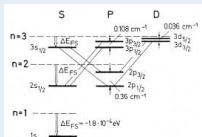


Numerical calculations

$$\frac{d}{dt} N_n^k = \text{fn} \left(N_{(m)}^{(i)}, N_e, T_h, T_e \right)$$

$$\frac{d}{dt} E_{e/h} = \text{fn} \left(N_{(m)}^{(i)}, N_e, T_h, T_e \right)$$

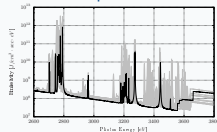
- ▶ Reduction capabilities for solver acceleration



Spectroscopic Tool

- ▶ Verified HEDP spectra from NLTE workshops
- ▶ Semiempirical stark broadening
- ▶ Continuum lowering for high density scenarios

Spectra



Challenges:

- ▶ Enumeration of atomic states and transitions (+ types of transitions!)
- ▶ Collisional-radiative effects are inherently connected to other physics modules

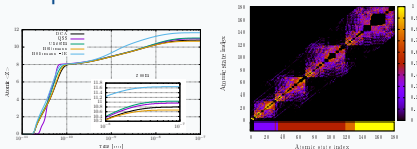
Research Direction

Focus: Plasma reaction kinetics

- ▶ Increase tractability of memory-intensive, multi-physics simulations
- ▶ Improve physical models and our understanding through plasma simulations

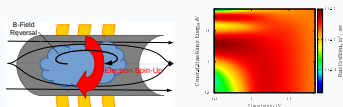
Reduced-order modeling

- ▶ Investigate complexity-reduction mechanisms to accelerate reaction models and preserve accuracy
- ▶ Machine-learning to minimize user intervention during reduction process

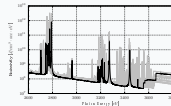


Higher fidelity modeling

- ▶ Relative multifluid streaming effect in simulating plasma reactions



- ▶ High-density, high-temperature spectroscopic comparisons





Outline

Introduction

Complexity-Reduction of Collisional-Radiative Models

Spectral Clustering of Atomic States for Reduced-Order Modeling

Development of Multifluid Collisional-Radiative Rate Coefficients

Conclusion



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Complexity-Reduction Strategies

Question:

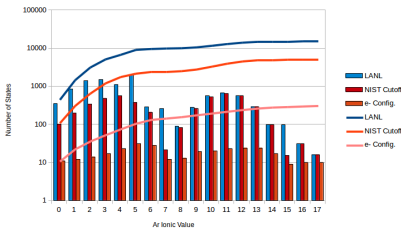
⇒ How can we expedite the reaction or collisional-radiative (CR) physics while maintaining consistency with the detailed simulated kinetics?

System of CR Equations

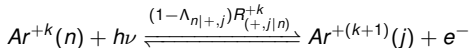
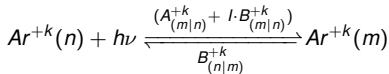
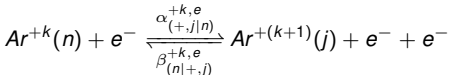
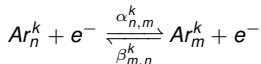
$$\frac{d}{dt} N_n^k = \text{fn} \left(N_{(m)}^{(i)}, N_e, T_h, T_e \right)$$

$$\frac{d}{dt} E_{e/h} = \text{fn} \left(N_{(m)}^{(i)}, N_e, T_h, T_e \right)$$

Argon Atomic Levels (LANL)



Atomic Processes Considered



Past Collisional-Radiative Reduction Strategies

Quasi-Steady-State (QSS) Solution

- ▶ Assumes fast kinetics between states within an ion distribution
- ▶ Also assumes longer diffusion/decay timescales than excited state lifetimes
- ▶ Must construct effective rate coefficients based on macroscopic parameters (N_e and T_e)

$$\frac{d}{dt} N_1^k = R^k - F^k \cdot N_1^k$$

$$\frac{d}{dt} N_n^k = 0 \quad \text{for } n > 1$$

⇒ Bates, Kingston, McWhirter (1962)

Uniform Grouping

- ▶ Multiple atomic groups are chosen within an ion distribution
- ▶ Individual levels and transition rates are weighted based on statistical weights within a group
- ▶ Rigid microscopic group description in atomic state distribution (ASD)

$$N_i = \mathcal{N}_n \frac{g_i}{g_n} \quad \text{where} \quad g_n = \sum_{i \in n} g_i$$

$$\tilde{\alpha}_{n,m} = \sum_{i \in n} \sum_{j \in m} \frac{g_i}{g_n} \alpha_{i,j}$$

Boltzmann Grouping Technique

Concept

- Construct an atomic group comprising of a base level, N_{n_0} , and excited subgroup, $\mathcal{N}'_n$
- Define a Boltzmann distribution within the group by using a group temperature description, T_n

Advantages

- Indirectly weighs levels and transitions based on conditions
- Flexible description of atomic groups for time-dependent simulations
- Better conservation properties for multiphysics simulations

⇒ Le, Karagozian, Cambier (2013)

Formulation

$$N_i = \frac{\mathcal{N}'_n}{\mathcal{Z}'_n} g_i e^{-\Delta E_i / T_n} \quad \text{where} \quad (1)$$

$$\mathcal{Z}'_n = \sum_{i \in n'} g_i e^{-\Delta E_i / T_n}$$

$$\tilde{\alpha}_{n',m} = \sum_{i \in n'} \sum_{j \in m} \frac{g_i e^{-\Delta E_i / T_n}}{\mathcal{Z}'_n} \alpha_{i,j} \quad (2)$$

- Requires group temperature (T_n) iterations to create best fit

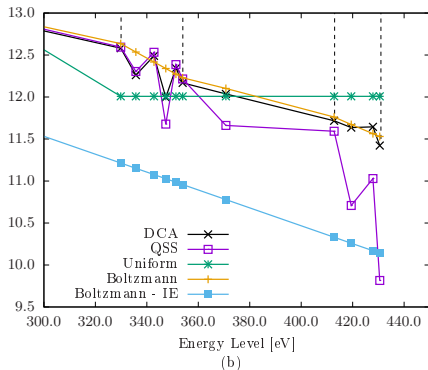
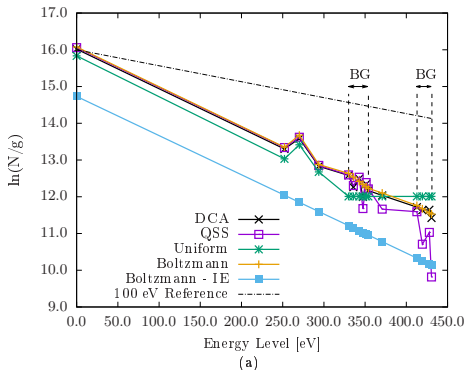
$$T_n^{(k)} - T_n^* = \frac{\ln \mathcal{Z}'_n(T_n^{(k)}) - \ln \mathcal{Z}'_n(T_n^*)}{\left[\frac{d \ln \mathcal{Z}'_n}{dT_n} \right] (T_n^*)} \quad (3)$$

$$\text{where } \mathcal{Z}'_n(T_n^{(k)}) = g_{n_0} \frac{\mathcal{N}'_n{}^{(k)}}{N_{n_0}^{(k)}}$$

Isothermal Heating Simulations

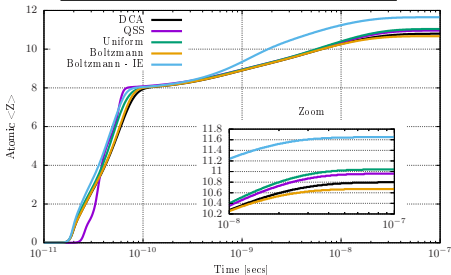
- ➔ Instantaneous change in electron energy to induce ASD evolution
- ➔ Explored with the detailed and complexity-reduced simulations

Boltzmann plot of Ar^{+8}

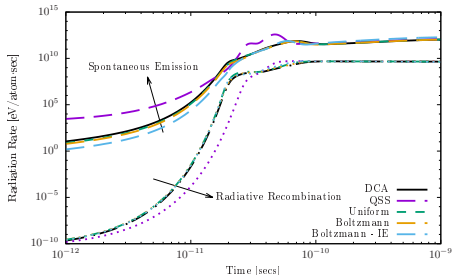


Collisional Case: $N_{Ar} = 10^{19} \text{ cm}^{-3}$, $T_e \rightarrow 100 \text{ eV}$

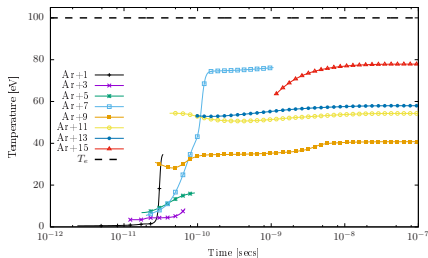
Group Temperature Evolution



Boltzmann Plot at Steady State

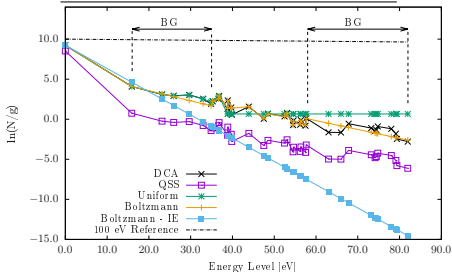


Group Temperature Evolution

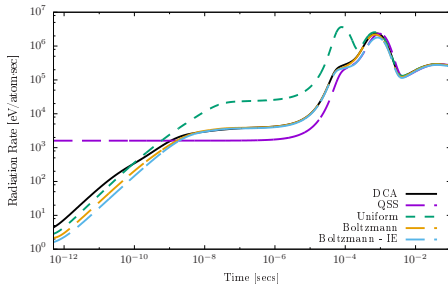


Coronal Case: $N_{Ar} = 10^{12} \text{ cm}^{-3}$, $T_e \rightarrow 100 \text{ eV}$

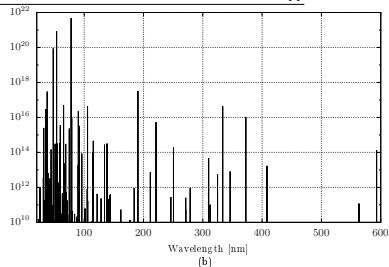
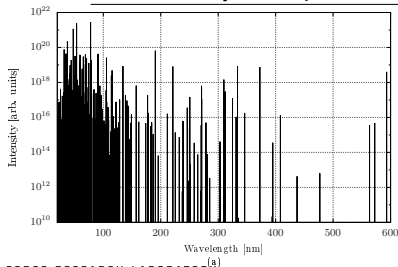
Group Temperature Evolution



Boltzmann Plot at Steady State



Radiation Spectra (DLA vs Boltzmann Ionic Excitation))



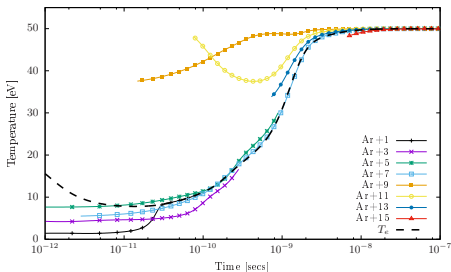


Planckian Irradiation Case - $T_{\text{rad}} = 50 \text{ eV}$

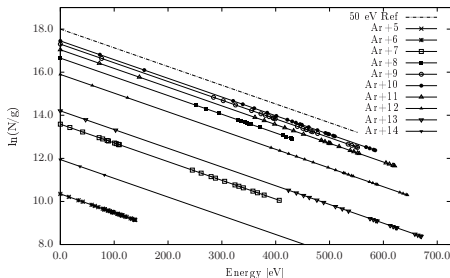
Objectives

- ⇒ Allow evolution of electron temperature, T_e , to effect changes in ASD
- ⇒ Verify that proper rates are determined from detailed balance / equilibrium

Group Temperature Evolution



Boltzmann Plot at Steady State





Quick Summary on Complexity-Reduction Strategies

- ▶ Generalized architecture for atomic collisional-radiative modeling with multiple complexity-reduction techniques
- ▶ Boltzmann grouping has several advantages as a reduction technique as shown in the isothermal and Planckian irradiation simulations

Method	Advantages	Disadvantages
Detailed Solution	• Most accurate model • Spectroscopic reconstruction	• Computationally expensive • Longest t-dep. simulations
QSS Solution	• Fastest t-dep. simulations	• Restrictive application regime
Boltzmann Grouping	• Flexible ASD representation • Conservative fluid properties	• Requires proper group selection



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Automatic Level Grouping

Following from the previous work...

- ▶ Question: Can machine learning algorithms automate the Boltzmann level grouping process for our reduced collisional-radiative model?

Procedure

1. Apply **unsupervised learning** algorithms to cluster atomic states
 - ▶ **Spectral Clustering**
2. Use the **Boltzmann grouping** technique devised in the previous section to describe these groups
3. Temporally evolve the reduced CR system of equations!

Why not supervised learning?

- ▶ Requires lots of training data $\therefore$ many sets of detailed CR simulations
- ▶ Parameter space is large and varies:
 - ▶ N_e , T_e , Plasma composition, Level completeness, Transition types, etc.
- ▶ Assignment of groupings for neural network learning can create more subjectivity

- ▶ Von Luxburg, U., A tutorial on spectral clustering, *Statistics and Computing*, 17, (4), (2007) 395-416.



Applying Spectral Clustering to Atomic States

Spectral Clustering Steps

1. Define *normalized* metric between states i & j :

$$S_{ij} = \frac{|\varepsilon_A^i - \varepsilon_A^j|^{\beta E}}{K_{i,j}^{\beta K}} \rightarrow W_{ij} = \exp\left(-\frac{S_{ij}}{\sqrt{\sigma_i \sigma_j}}\right)$$

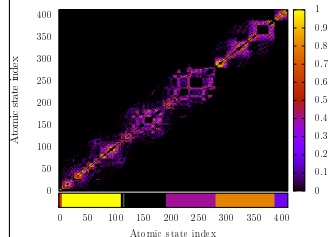
2. Calculate asymmetric normalized graph Laplacian:

$$L_{rw} = I - D^{-1}W, \quad D(i, j) : D_{ij} = \begin{cases} \sum_{j=i}^n W_{ij}, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases}$$

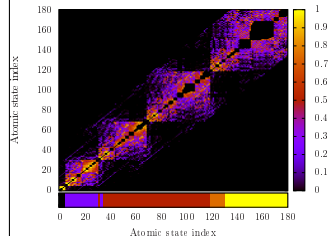
3. Extract N_{grp} smallest eigenvalues (+ eigenvectors) to execute chosen clustering technique
4. Perform K-medoids clustering on extracted eigenvectors and transfer groupings to CR model

Constructed Groups

O⁺¹ Graph and Groupings



O⁺² Graph and Groupings





Simulation Setup

Objectives:

- ⇒ Compare detailed-level-accounting (DLA) results with clustering simulations

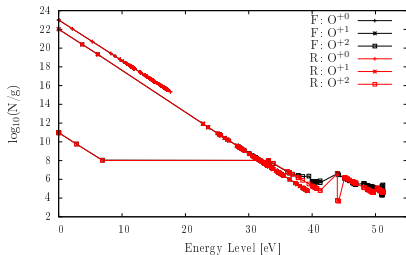
Atomic & Transition Data

- ▶ Atomic Model: cFAC
- ▶ Atomic Species: O^{+0} to O^{+2}
 - ▶ *jj-coupling description*
- ▶ Transition Types:
 - ▶ e^- Excitation*
 - ▶ e^- Ionization
 - ▶ Spontaneous Emissions*
 - ▶ Radiative Recombination
- ▶ Considered electric and magnetic dipole + quadrupole transitions for thorough connection matrix

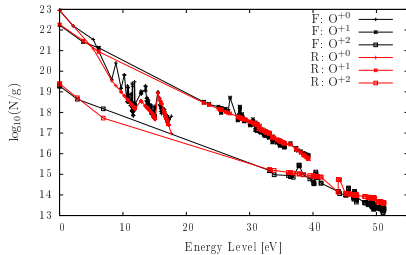
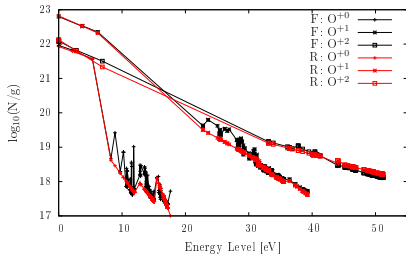
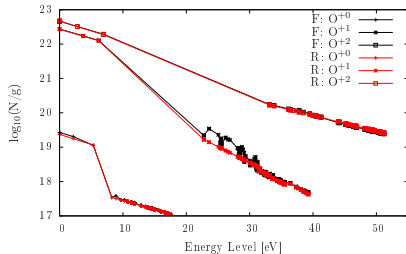
Atomic Modeling Statistics		
Ion	# DLA	# Groups
O^{+0}	549	7
O^{+1}	413	6
O^{+2}	180	5

Ion	Excitation Channels	Emission Lines
O^{+0}	148,733	173,838
O^{+1}	84,029	103,835
O^{+2}	15,912	19,249

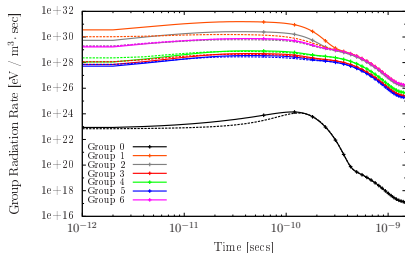
Collisional Case: $N_{Ar} = 10^{18} \text{ cm}^{-3}$, $T_e = 1 \rightarrow 10 \text{ eV}$



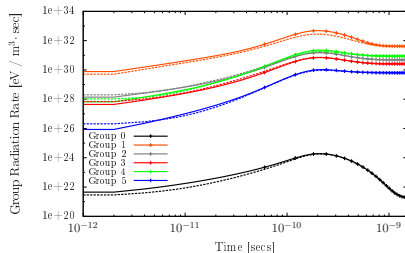
(a) Initial


(b) $4.01 \times 10^{-11} \text{ secs}$

(c) $1.82 \times 10^{-10} \text{ secs}$

(d) $3.25 \times 10^{-10} \text{ secs}$

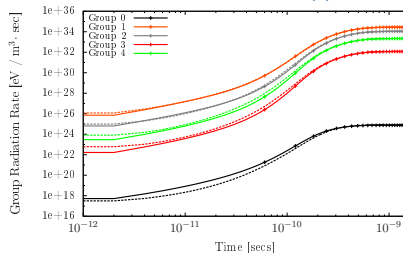
Collisional Case: $N_{Ar} = 10^{18} \text{ cm}^{-3}$, $T_e = 1 \rightarrow 10 \text{ eV}$



(a) O⁺0 Group Radiation Loss

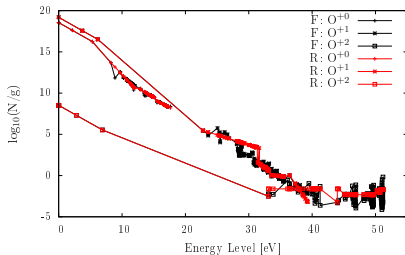


(b) O⁺1 Group Radiation Loss

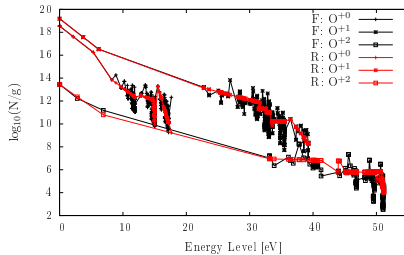


(c) O⁺2 Group Radiation Loss

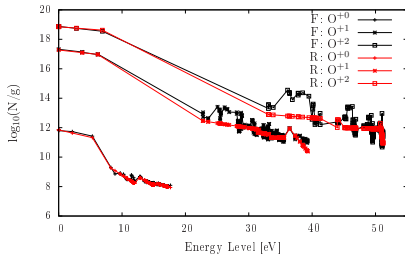
Coronal Case: $N_{Ar} = 10^{14} \text{ cm}^{-3}$, $T_e = 1 \rightarrow 10 \text{ eV}$



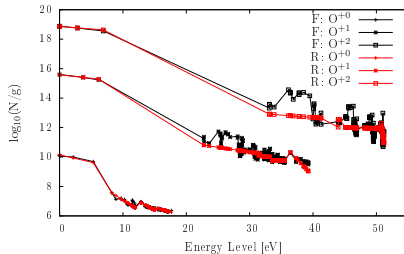
(a) Initial



(b) $1.45 \times 10^{-10} \text{ secs}$

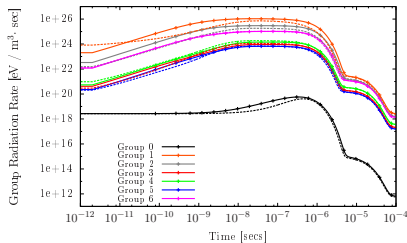


(c) $3.40 \times 10^{-5} \text{ secs}$

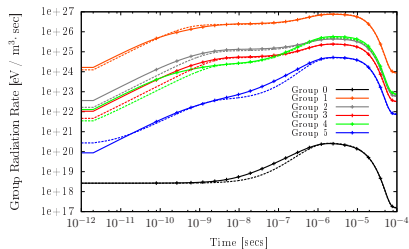


(d) $1.00 \times 10^{-4} \text{ secs}$

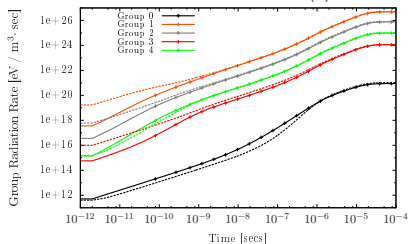
Coronal Case: $N_{Ar} = 10^{14} \text{ cm}^{-3}$, $T_e = 1 \rightarrow 10 \text{ eV}$



(a) O^{+0} Group Radiation Loss



(b) O^{+1} Group Radiation Loss



(c) O^{+2} Group Radiation Loss



Quick Summary on Spectral-Clustered Atomic Groupings

Wall-clock Timing			
	<i>Coronal</i>	<i>Collisional</i>	<i># of Vars.</i>
<i>Full</i>	2298 secs	2030 secs	1143
<i>Reduced</i>	331 secs	287	36
<i>Speedup</i>	6.94×	7.07×	31.8×

- ▶ Demonstrated feasibility of applying spectral clustering to a collisional-radiative model employing Boltzmann grouping
- ▶ Spectral clustering the selection of atomic level groups; demonstrated in this work using detailed-level-accounting (DLA) states
- ▶ Exercise caution when encountering different plasma regimes (e.g. collisional vs. coronal conditions)



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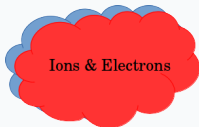


Multifluid Concept behind Rate Calculation

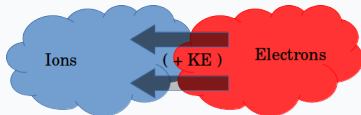
Objective:

- ⇒ Exploit the additional energy carried by a component of the multifluid plasma for collisional-radiative processes / calculations

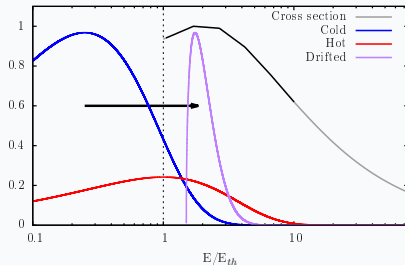
Conventional Rate Calculation



Rates with Streaming



Impact of Streaming on Rate Calculation





Multifluid Rate Equations

Generalized Transfer Integral : $\Psi_{si} = N_s N_i \int d^3 \mathbf{v}_s d^3 \mathbf{v}_i f_s f_i g \int \psi d\sigma(\mathbf{v}_s, \mathbf{v}_i; \mathbf{v}_t, \mathbf{k})$

Electron-impact Excitation

$$\Gamma_{lu} = N_e N_l \bar{g}_{\bar{T}} e^{-\lambda} \int_{x=x^*}^{\infty} dx x e^{-x} \bar{\sigma}_{lu}(x) \times \zeta^{(0)}(\sqrt{\lambda x})$$

Electron-impact Deexcitation

$$\Gamma_{ul} = N_e N_u \bar{g}_{\bar{T}} e^{-\lambda} e^{x^*} \frac{g_l}{g_u} \int_{x=x^*}^{\infty} dx x e^{-x} \bar{\sigma}_{lu}^{ex}(x) \times \zeta^{(0)}(\sqrt{\lambda(x - x^*)})$$

Moment Integration

- ▶ 0th ($\psi = 1$): Density
- ▶ 1st ($\psi = m\mathbf{v}$): Momentum
- ▶ 2nd ($\psi = 0.5m\mathbf{v}^2$): Energy

Applied to Other Atomic Processes

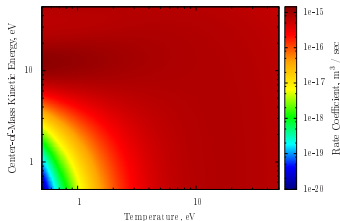
- ▶ Electron-impact Ionization
- ▶ Three-body Recombination
- ▶ Radiative Recombination

- ▶ Le, H. P., and Cambier, J. L. "Modeling of inelastic collisions in a multifluid plasma: Excitation and deexcitation." *Physics of Plasmas*, 22(9), 093512, 2015.
- ▶ Le, H. P., and Cambier, J. L. "Modeling of inelastic collisions in a multifluid plasma: Ionization and recombination", *Physics of Plasmas*, 23(6), 063505, 2016.

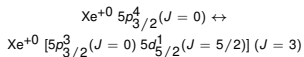
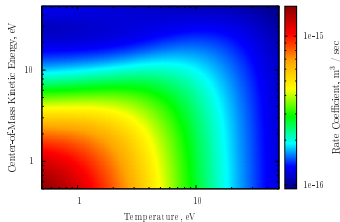
Sample Multifluid Rates: Excitation/De-excitation

Allowed Transition

Excitation

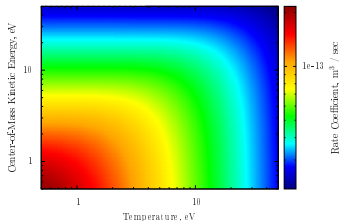


Deexcitation

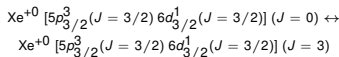
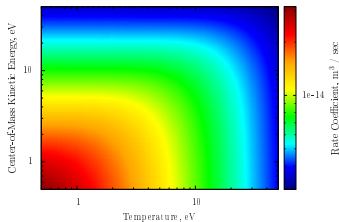


Spin-Forbidden Transition

Excitation

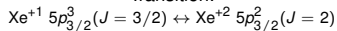


Deexcitation

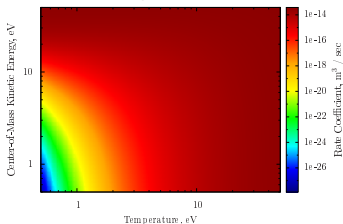


Other Multifluid Rate Coefficient Maps

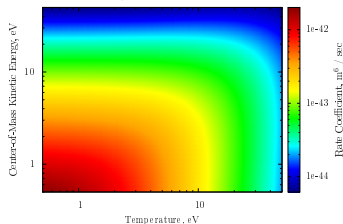
Transition:



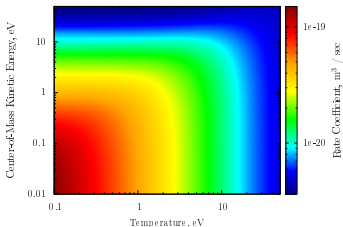
e^- -impact Ionization



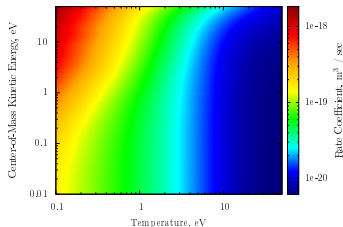
e^- -impact Recombination



Rad. Recomb. Density Rate

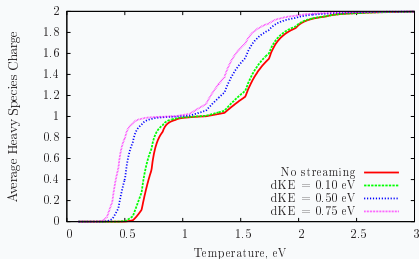


Rad. Recomb. Energy Rate

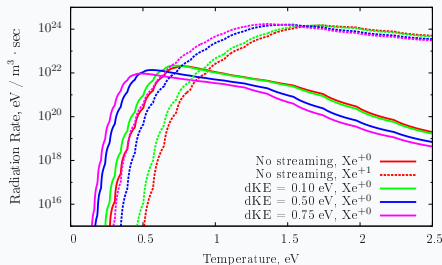


Streaming Effects on Steady State Parameters

Average Charge



Radiation Rate



Lessons:

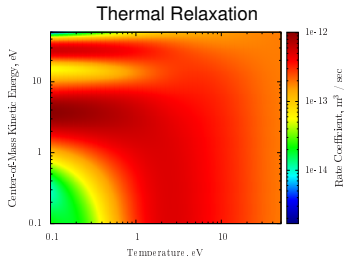
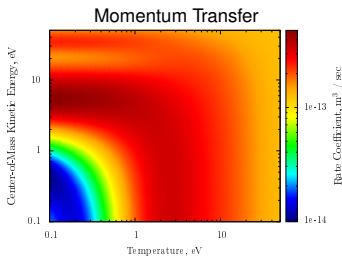
- ▶ Increasing relative kinetic energy leads to higher $\bar{Z}$
- ▶ Higher radiation loss rates resulting from population of higher excited states

Multifluid Xe⁺⁰ Elastic Collision Rates¹

Electron-Neutral

$$K^{en} = m_e N_e N_{neu} \frac{2}{3} \bar{g}_{\tilde{T}} e^{-\lambda} \int_{x=x^*}^{\infty} dx x^2 e^{-x} \bar{\sigma}^{en}(x) (1 - \langle \cos \chi \rangle_{\Omega'}) \zeta^{(1)}(\sqrt{\lambda x})$$

$$J^{en} = N_e N_{neu} \bar{g}_{\tilde{T}} e^{-\lambda} \int_{x=x^*}^{\infty} dx x^2 e^{-x} \bar{\sigma}^{en}(x) (1 - \langle \cos \chi \rangle_{\Omega'}) \left[\zeta^{(0)}(\sqrt{\lambda x}) - \frac{2\lambda}{3} \zeta^{(1)}(\sqrt{\lambda x}) \right]$$



⇒ **Ramsauer-Townsend effect** in x-sections translates onto rate coefficient maps! ⇐

¹McEachran, R. P., and Stauffer, A. D., The European Physical Journal D, 68(6), 153, 2014.



0-D Collisional-Radiative Equations

$$\frac{dN_i^{+k}}{dt} = \Gamma_{+k,i}^{(inelas)} \quad \text{for all ions, } +k, \text{ and all respective states, } i \quad (4)$$

$$\frac{d(m_e N_e \mathbf{u}_e)}{dt} = \mp (\mathbf{u}_e - \mathbf{u}_h) \left(K^{(ec,neu)} + K^{(ec,ion)} + K^{(inelas)} \right) \quad (5)$$

$$\begin{aligned} \frac{d\varepsilon_e}{dt} = & \mp (\bar{\mathbf{u}} - \mathbf{u}_e) \cdot (\mathbf{u}_e - \mathbf{u}_h) \left(K^{(ec,neu)} + K^{(ec,ion)} + K^{(inelas)} \right) \\ & \mp 2 \frac{m_e}{m_h} k (T_e - T_h) \left(J^{(ec,neu)} + J^{(ec,ion)} + J^{(inelas)} \right) - \left[\Xi^{(inelas)} \right]_{(e^- \text{ Eqn})} \end{aligned} \quad (6)$$

Absent Physical Phenomena:

1. Spatial effects, e.g. particle / fluid convection
2. Momentum equation + Diffusion processes + Fluid resistivity
3. Convective heat flux / Energy transport by particles
4. Momentum transfer / Thermal relaxation terms from inelastic collisions



Simulated Multifluid Plasma Evolution

Task:

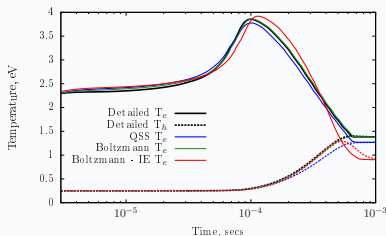
⇒ Simulate ionization of stationary atomic particles caused by streaming electrons

Modeling Setup

- ▶ $N_{\text{heavy}} = 10^{13} \text{ cm}^{-3}$
- ▶ $T_{e,\text{init}} = 0.65 \text{ eV}$, $T_{h,\text{init}} = 0.25 \text{ eV}$
- ▶ Electron seeding for initialization
- ▶ Detailed and Reduced Configurations
- ▶ Ionization Levels:
 Xe^{+0} (53), Xe^{+1} (117), Xe^{+2} (5)
- ▶ Number of Transitions:
8166 Excitation, 4150 Ionization,
2600 Lines

CR Forcing

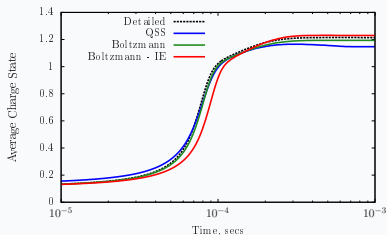
$$dKE(t) = \begin{cases} 0.05 \text{ eV} \cdot \text{erf}(t / 0.1 \mu\text{secs}) & \text{if } Z < \frac{1}{2} \\ dKE(t_{Z=\frac{1}{2}}) \cdot \exp(t_{Z=\frac{1}{2}} - t) & \text{for } t > t_{Z=\frac{1}{2}} \end{cases}$$



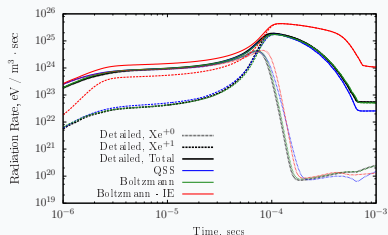
- ▶ Complete set of atomic and transition data were generated using cFAC

Evolution of 0-D Multifluid Rate Simulation

Reduced-order modeling



Higher fidelity modeling





Quick Summary on Multifluid Rate Coefficients

- ▶ Derived multifluid rate coefficients for radiative recombination processes
- ▶ Added capability in CR model to account for multifluid rate coefficients
 - ▶ Electron-impact excitation and deexcitation
 - ▶ Electron-impact ionization and 3-body recombination
 - ▶ Radiative Recombination
- ▶ Demonstrated significance of multifluid rate coefficient effects for reactive, multifluid plasma systems
- ▶ Also applied complexity-reduction techniques to CR test case with multifluid rate coefficients



Outline

Introduction

Complexity-Reduction of Collisional-Radiative Models

Spectral Clustering of Atomic States for Reduced-Order Modeling

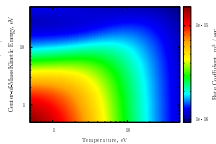
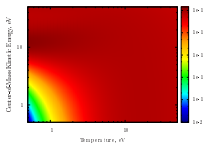
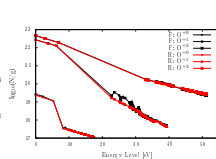
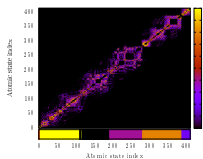
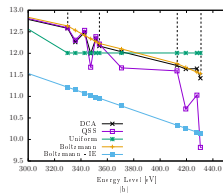
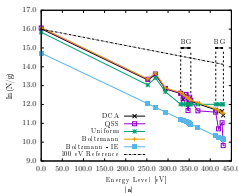
Development of Multifluid Collisional-Radiative Rate Coefficients

Conclusion



Final Thoughts

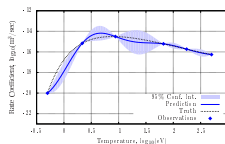
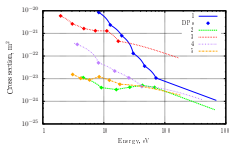
- ▶ Developed a collisional-radiative model with complexity-reduction capabilities: quasi-steady-state (QSS) solution, uniform grouping, and Boltzmann grouping
- ▶ Applied spectral clustering to minimize user intervention during construction of level groups for reduced CR simulations
- ▶ Extended multifluid rate coefficient formulation to radiative recombination processes and added multifluid rate capabilities to the model



Ongoing Tasks

Rate Coefficient Regression:

- ▶ Exploring capabilities to reduce memory footprint of collisional-radiative rate coefficients for time-dependent applications
- ▶ Applied Gaussian process (GP) regression to conventional T_e -based coefficients; planning extension to multifluid rate coefficient ($T_e, \Delta KE$)



Going beyond 0-D Simulations:

- ▶ Coupling collisional-radiative model to radiation transport equation
- ▶ Developing complexity-reduction capabilities involving both physics modules

$$\left[\frac{1}{c} \frac{\partial}{\partial t} + \frac{\partial}{\partial s} \right] I(\mathbf{x}, t, \mathbf{n}, \nu) = \eta(\mathbf{x}, t, \mathbf{n}, \nu) - \chi(\mathbf{x}, t, \mathbf{n}, \nu) I(\mathbf{x}, t, \mathbf{n}, \nu)$$



Author's Publications

- ▶ Abrantes, R. J. E., A. R. Karagozian, D. Bilyeu, and H. P. Le. "Complexity reduction effects on transient, atomic plasmas." *Journal of Quantitative Spectroscopy and Radiative Transfer* 216 (2018): 47-55.
- ▶ Abrantes, R. J. E., È. Sousa, D. Bilyeu, and R. Martin. "Complexity-reduction using automatic level grouping for atomic collisional-radiative models." *Journal of Computational Physics* 407 (2020): 109213.
- ▶ Abrantes, R. J. E., and R. S. Martin. "Reduced-order modeling of plasma ionization due to multifluid, collisional-radiative effects." *Physics of Plasmas* 28, No. 2 (2021): 022104.
- ▶ Abrantes, R. J. E., Y.-W. Mao, D. D. W. Ren, "Rate coefficient function estimation using Gaussian process regression", *In Preparation*.



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Questions?